Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

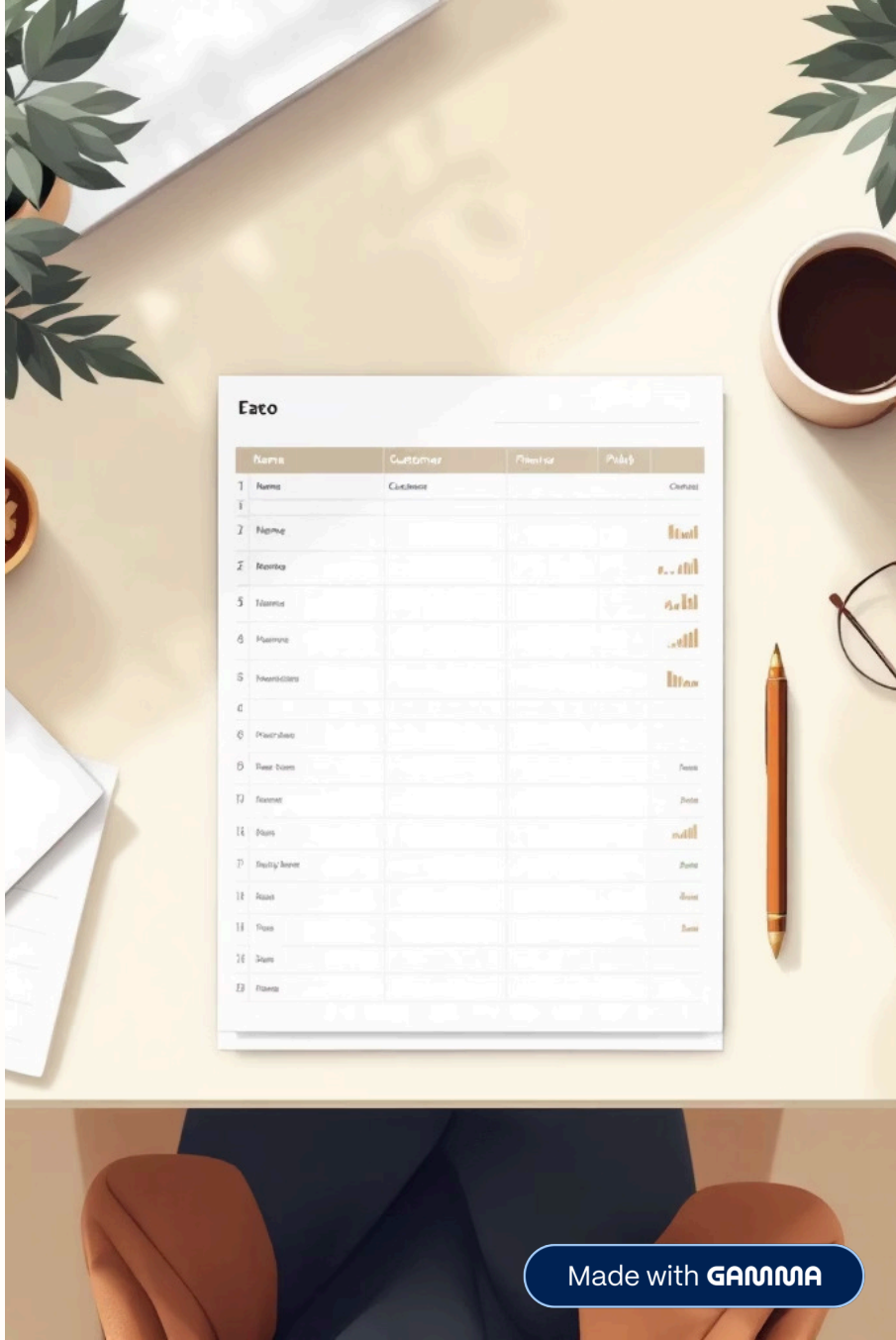
Data Columns

Customer and purchase details

37

Missing Values

Review ratings imputed



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Data Preparation Process

Data Loading

Imported dataset using pandas

Initial Exploration

Checked structure and summary statistics

Missing Data Handling

Imputed using median ratings by category

Feature Engineering

Created age groups and purchase frequency

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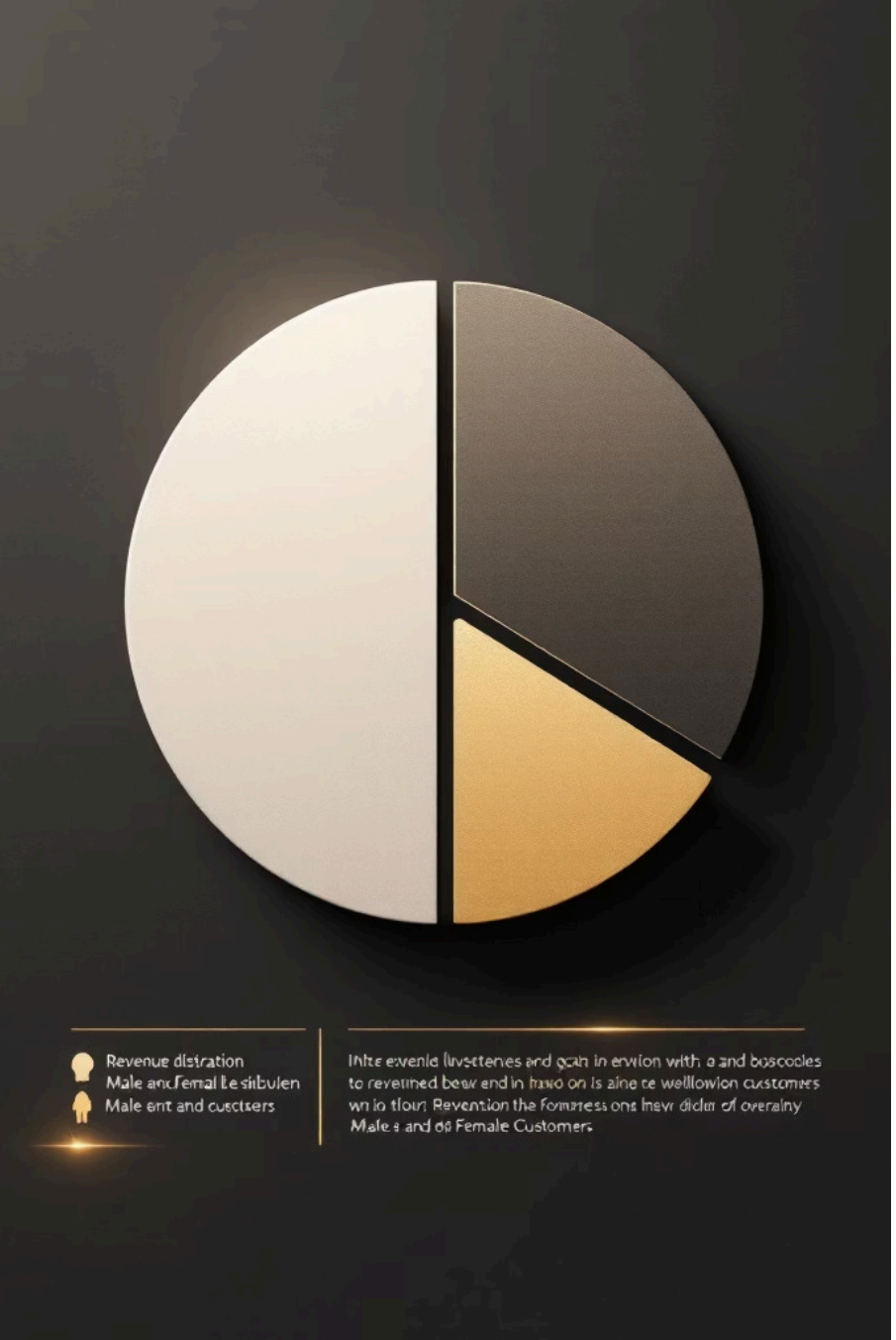

SQL Analysis: Revenue Insights

Revenue by Gender

Compared total revenue generated by male vs. female customers

Shipping Comparison

Standard vs. Express shipping average purchase amounts

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SQL Analysis: Customer Segments

Top Products by Rating

Products with highest average review ratings

Discount-Dependent Products

5 products with highest percentage of discounted purchases

Customer Segmentation

Classified as New, Returning, or Loyal based on purchase history

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SQL Analysis: Subscription Behavior

Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status

Repeat Buyers & Subscriptions

Customers with >5 purchases more likely to subscribe

High-Spending Discount Users

Identified customers who used discounts but spent above average

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SQL Analysis: Product Performance

Top 3 Products per Category

Most purchased products within each category

High-Rated Products

Products with highest average review ratings

Discount-Dependent Products

Products with highest percentage of discounted purchases

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Power BI Dashboard

Interactive visualizations presenting key insights

Revenue by Age Group

Total revenue contribution of each age group

Geographic Analysis

Customer location and regional spending patterns

Seasonal Trends

Purchase patterns across different seasons

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Customer Loyalty Programs

Reward repeat buyers to move into "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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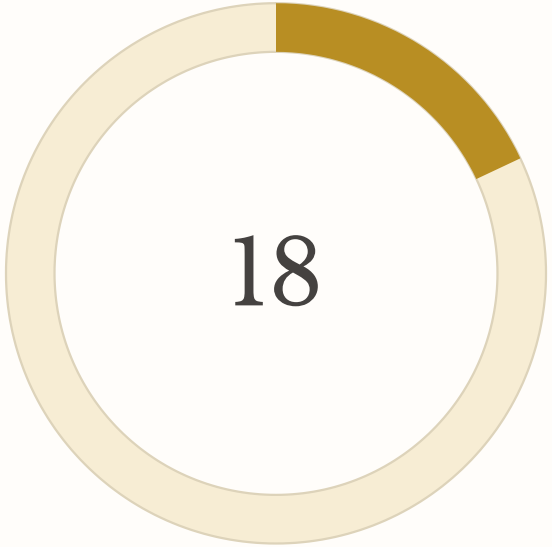
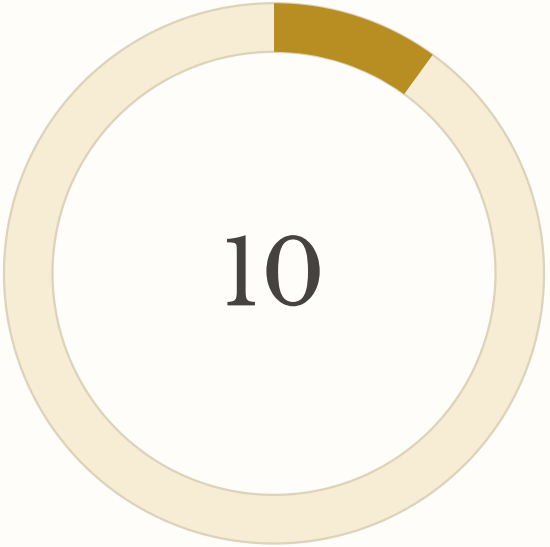
Key Takeaways

Purchase Records Analyzed



Data Dimensions



SQL Queries Run

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